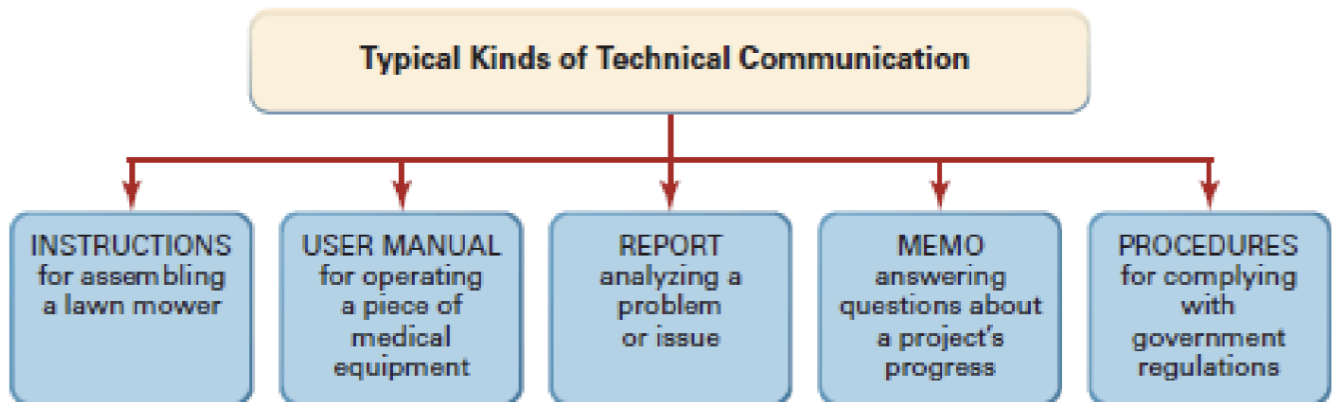


Unit 1



Main Features Of Technical Communication:

1. Reader-Centered
2. Often Produced by Teams
3. Delivered in Paper and Digital versions

Main Features Of Technical Communication

- Almost any form of technical communication displays certain shared features:
- The communication is reader-centered, accessible and efficient, often produced by teams, and delivered in both paper and digital versions.
- **Reader-Centered**
 - Unlike poetry, fiction, or college essays, a technical document rarely focuses on the writer's personal thoughts and feelings.
 - This doesn't mean that your document should have no personality (or voice), but it does mean that the needs of your readers come first.
 - Workplace readers typically are interested in "who you are" only to the extent that they want to know what you have done, what you recommend, or how you speak for your company.
 - Reader-centered documents focus on what people need to learn, do or decide.

- Accessible and Efficient
- Readers expect to find the information they need and to have questions answered clearly. For instance, the document shown in Figure is written and designed so that a nontechnical audience can find and follow the information.
- Instead of long technical passages, the content is presented in short chunks, answering the main question readers will ask (how to choose the right model).

What is Technical Writing?

- **Definition:** Technical writing is the process of documenting, explaining, and communicating complex technical information clearly, concisely, and accurately to a specific audience.
- **Key Characteristics:**
 - **Clarity:** The information must be easy to understand.
 - **Accuracy:** The information must be factually correct.
 - **Conciseness:** Avoids unnecessary words and jargon.
 - **Audience-Oriented:** Tailored to the knowledge level and needs of the reader.
 - **Purposeful:** Serves a specific goal (e.g., to instruct, to inform, to persuade).

Role #1 : Documenting the Design & Process

- Engineering Notebooks & Logbooks
- Design Specifications & Requirements
- Test Plans & Reports

Role #2: Communicating within the Team & Company

- Internal Memos & Emails
- Project Proposals & Progress Reports
- Presentations & Slide Decks

Role #3: Communicating with External Stakeholders

- User Manuals & Instructions
- Client Reports
- Academic Papers & Conference Proceedings

The four pillars of effective technical communication:

- Clarity - The quality of being easy to understand. Your message should be unmistakable.
- Conciseness - Saying what needs to be said in the fewest possible words, without sacrificing clarity. Time is a valuable resource
- Accuracy - The quality of being correct and free from errors. In engineering, mistakes can have serious consequences
- Audience Focus - Tailoring your communication to the knowledge level, needs, and expectations of your reader

Technical communication process:

1. Planning: The foundation of your document.

- Brainstorm
- Outline
- Gather Resources

2. Drafting: Getting the ideas on paper.

- Follow Your Outline
- Turn off your inner editor
- Write in chunks
- Focus on the data

3. Revising: Restructuring and refining the content.

- Clarity
- Logical Flow
- Audience Focus
- Completeness
- Methods for Effective Revision:
 - Step Away
 - Read Aloud

- Seek Feedback

4. Editing: Polishing for clarity and correctness.

- Grammar & Punctuation
- Spelling
- Formatting
- Numbers & Units

Summary

- Planning: Define audience, purpose, scope, and create an outline.
- Drafting: Focus on getting ideas down without worrying about perfection.
- Revising: Check the big picture for clarity, flow, and audience focus.
- Editing: Fine-tune the details for grammar, spelling, and consistency.

The takeaway: This systematic process transforms technical writing from a daunting task into a manageable series of steps. It allows you to produce high-quality, professional documents that reflect the quality of your engineering work.

The Three Tiers of Audience:

1. Primary audience → main target audience, direct recipients
2. Secondary audience → reviewers or advisors
3. Tertiary audience → broader, more distant group

Primary and secondary Audiences

- Generally, primary readers are decision makers who requested the document.
- Secondary readers are those who will carry out the project, who will advise the decision makers, or who will be affected by this decision in some way.

Guidelines For Deciding About Tone

- **Use a formal or semiformal tone** in writing for superiors, professionals, or academics (depending on what you think the reader expects).
- **Use a semiformal or informal tone** in writing for colleagues and subordinates (depending on how close you feel to your reader).
- **Use an informal tone** when you want your writing to be conversational, or when you want it to sound like a person talking. But always be professional.
- **Avoid a negative tone** when conveying unpleasant information.
- **Above all, find out what tone your particular readers prefer.** When in doubt, do not be too casual!

Purposes Of Technical Communication

1. to anticipate and answer questions (inform your readers)
2. to enable people to perform a task or follow a procedure (instruct your readers)
3. to influence people's thinking (persuade your readers)

Guidelines For Nonsexist Usage

- **Use neutral expressions** such as chair or chairperson rather than chairman and postal worker rather than postman.
- **Rephrase to eliminate the pronoun**, but only if you can do so without altering your original meaning. For instance, change A writer will succeed if he revises to A writer who revises succeeds.
- **Use plural forms** such as Writers will succeed if they revise (but not A writer will succeed if they revise). For pronoun-referent agreement, see page 667.
- **Use occasional paired pronouns** (him or her, she or he, his or hers): A writer will succeed if she or he revises.
- **Drop condescending diminutive endings** such as -ess and -ette used to denote females (poetess, drum majorette, actress).

Producing an effective document typically requires that you complete the four basic tasks:

1. Information Delivery

2. Good Teamwork
 3. Ethical Presentation
 4. Persuasive Reasoning
-

Unit 2

Procedural Stages of the Research Process:

1. Searching for information
2. Record your findings
3. Documenting your sources
4. Writing the document

Stages of Critical Thinking in the Research Process:

1. Asking the right questions
2. Exploring a balance of views → Approach your research topic from a variety of angles
3. Achieving adequate depth in your research
4. Evaluate your findings
5. Interpret your findings

Questions for evaluating a particular finding:

1. Is this information accurate, reliable, and relatively unbiased?
2. Do the facts verify the claim?
3. How much of the information is useful?
4. Is this the whole or the real story?
5. Do I need more information?

Questions for interpreting your findings:

1. What are my conclusions and do they address my original research question?
2. Do any findings conflict?
3. Are other interpretations possible?
4. Should I reconsider the evidence?
5. What, if anything, should be done?

Secondary Sources:

- **Subject directories** are indexes compiled by editors who sift through Web sites and sort the most useful links.
 - eg: About.com
- Some of the most popular general **periodical databases** include InfoTrac, NewsBank, ProQuest and EBSCOHost
- **Webmaster** (the person who created and maintains the site)

Guidelines For Researching On The Internet

1. Expect limited results from any one search engine or subject directory
2. When using a search engine, select keywords or search phrases that are varied and technical rather than general
3. When using Wikipedia or other online encyclopedias, check out the footnotes and other citations
4. Consider the domain type (where the site originates)
5. Identify the site's purpose and sponsor
6. Look beyond the style of a site
7. Assess the currency of the site's materials
8. Assess the author's credentials and assertions
9. Use bookmarks and hotlists for quick access to favorite Web sites
10. Save or print what you need before it changes or disappears
11. Download only what you need; use it ethically; obtain permission; and credit your sources

Traditional Secondary Sources

1. Books and Periodicals
2. Reference works (general information sources that provide background and can lead to more specific information)
3. Bibliographies (lists of books and/or articles categorized by subject)
4. Indexes (Book and article bibliographies)
 - eg: Index to Scientific and Technical Proceedings
 - eg: Newspaper indexes- New York Times
5. Periodical indexes (These indexes list articles from magazines and journals)
 - eg: Readers' Guide to Periodical Literature
6. Citation indexes (Using a citation index, you can track down the publications in which original material has been cited, quoted, or verified)

7. Technical report indexes (government and private-sector reports)
 - eg: Scientific and Technical Aerospace Reports index
8. Patent indexes (These indexes list articles from magazines and journals)
9. Encyclopedias (Encyclopedias are alphabetically arranged collections of articles)
 - eg: Encyclopedia Britannica - most popular general encyclopedia
10. Dictionaries (alphabetically arranged lists of words, including definitions, pronunciations, and word origins)
11. Handbooks (condensed facts (formulas, tables, advice, examples) about particular fields)
12. Almanacs (collections of factual and statistical data, usually arranged by subject area and published annually)
13. Directories (updated information about organizations, companies, people, products, services, or careers, often listing addresses and phone numbers)
 - eg: The Career Guide: Dun's Employment Opportunities Directory
 - eg: Directory of American Firms Operating in Foreign Countries
14. Abstracts (collected summaries of books and/or articles)
15. Gray literature (Some useful printed information may be unavailable at any library. This is known as "gray literature," or materials that are unpublished or not typically catalogued)
 - eg: pamphlets, dissertations

Primary Sources (interviews, surveys, and other techniques)

1. Unsolicited Inquiries
2. Informational Interviews
3. Surveys
 - While interviews allow for greater clarity and depth, surveys offer an inexpensive way to get the viewpoints of a large group

Evaluate the Sources

- Evaluate your sources based on currency, reliability, and other factors
1. Determine the currency of the source
 2. Assess the reputation of the source
 3. Consider the possible motives of those who have funded the study
 4. Cross-check the source against other, similar sources

Evaluate the Evidence

1. Determine the sufficiency of the evidence
2. Differentiate hard from soft evidence
 - “Hard evidence” consists of facts, expert opinions, or statistics that can be verified.
 - “Soft evidence” consists of uninformed opinions or speculations, data that were obtained or analyzed unscientifically, and findings that have not been replicated or reviewed by experts.
3. Decide whether the presentation of evidence is balanced and reasonable
4. Consider how the facts are being framed

Interpret Your Findings

Interpreting means trying to reach the truth of the matter

1. Identify your level of certainty
 - a. The ultimate truth—the conclusive answer
 - b. The probable answer
 - c. The inconclusive answer

Examine The Underlying Assumptions

Avoid Distorted Or Unethical Reasoning

Avoid Distorted Or Unethical

- Causes for Distorted Reasoning:
 - **Faulty Generalization**
 - **Faulty Causal Reasoning**
 - **Faulty Statistical Analysis**
 - determine the meaning of a collected set of numbers

The Meaningless Statistic

The Undefined Average

The Distorted Percentage Figure

The Bogus Ranking

Confusion Of Correlation With Causation

- **Correlation** is a numerical measure of the strength of the relationship between two variables
- **Causation** is the demonstrable production of a specific effect
- **confounding factors** - which are other possible reasons or explanations for a particular outcome

The Biased Meta-analysis

The Fallible Computer Model

Misleading Terminology

Acknowledge The Limits Of Research:

1. Obstacles To Validity And Reliability
 - A survey, is valid when:
 - (1) it measures what you want it to measure
 - (2) it measures accurately and precisely
 - (3) its findings can be generalized to the target population.
2. Deceptive Reporting

Skimming Books and Articles

- In a book skim:
 1. The preface and introduction
 2. The acknowledgments section
 3. The table of contents
 4. The notes at the ends of chapters or at the end of the book
 5. The index
 6. A few paragraphs from different portions of the text
- In an article skim:

1. The abstract
2. The introduction
3. The notes and references
4. The headings and several of the paragraphs

Recording Bibliographic Information

- Information to record for a book:
 1. author
 2. title
 3. publisher
 4. place of publication
 5. year of publication
 6. call number or URL
- Information to record for an article:
 1. author
 2. title of article
 3. title of periodical
 4. volume
 5. number
 6. date of publication
 7. pages on which article appears
 8. call number or URL of periodical

Paraphrasing Accurately

1. Study the original until you understand it thoroughly
2. Rewrite the relevant portions of the original
3. Title the information so that you'll be able to identify its subject at a glance
4. Include the author's last name, a short title of the article or book, and the page number (if any) of the original

Summarizing

1. Read the passage carefully several times
2. Underline key ideas
3. Combine key ideas
4. Check your draft against the original for accuracy and emphasis
5. Record the bibliographic information carefully

Documenting Your Sources

1. It helps you acknowledge your debt to your sources
2. It helps you establish credibility
- Three kinds of material should always be documented:
 - Any quotation from a written source or an interview, even if it is only a few words
 - A paraphrased idea, concept, or opinion gathered from your reading
 - Any graphic from a written or an electronic source

APA Style (American Psychological Association)

- citations in the text and a list of references at the end of the document.

APA Style for Textual Citations

1. Summarized or Paraphrased Material
 - include the author's name and the publication date in parentheses immediately following the borrowed information
 - If your sentence already includes the source's name, do not repeat it in the parenthetical notation
2. Quoted Material or Specific Fact
 - If the reference is to a specific fact, idea, or quotation, add the page number(s) from the source to your citation
3. Source with Multiple Authors
 - Use an ampersand (&) in the parenthetical citation itself, but use the word and in regular text
4. Source Authored by an Organization
 - use the name of the organization
 - If the organization name is commonly abbreviated, you may include the abbreviation in the first citation and use it in any subsequent citations
5. Source with an Unknown Author
 - use a shortened version of the title in your parenthetical citation
 - If the author is identified as anonymous—a rare occurrence—treat Anonymous as a real name
6. Multiple Authors with the Same Last Name
 - Use first initials if two or more sources have authors with the same last name
7. Multiple Sources in One Citation

- present the sources in alphabetical order, separated by a semicolon

8. Personal Communication

- When you cite personal interviews, phone calls, letters, memos, and emails, include the words personal communication and the date of the communication

9. Electronic Document

- Cite the author and date for an electronic source as you would for other kinds of documents
- If the author is unknown, give a shortened version of the title in your parenthetical citation
- If the date is unknown, use n.d
- If the document is posted as a PDF file, include the page number in the citation
- If a page number is not available but the source contains paragraph numbers, give the paragraph number
- If no paragraph or page number is available and the source has headings, cite the appropriate heading and paragraph

APA Reference List

1. Arranging entries

- Arrange the entries alphabetically by author's last name. If two or more works are by the same author, arrange them by date, earliest to latest. If two or more works are by the same author in the same year, list them alphabetically by title and include a lowercase letter after the date: 2010a, 2010b, and so on

2. Book titles

- Italicize titles of books. Capitalize only the first word of the book's title, the first word of the subtitle, and any proper nouns

3. Publication information

- For books, give the publisher's name in as brief a form as is intelligible; retain the words Books and Press. Include the name of both the city and the state (abbreviated) for publishers located in U.S. cities or the city and the country (not abbreviated) for publishers in non-U.S. cities; for publishers located in Canadian cities, also include the province

4. Periodical titles

- Italicize titles of periodicals and capitalize all major words

5. Article titles

- Do not italicize titles of articles or place them in quotation marks. Capitalize only the first word of the article's title and subtitle and any proper nouns

6. Electronic sources

- Include as much information as you can about electronic sources, such as author, date of publication, identifying numbers, and retrieval information. Include the digital object identifier (DOI) when one exists

7. Indenting

- Use a hanging indent, with the first line of each entry flush with the left margin and all subsequent lines indented one-half inch

8. Spacing

- Double-space the entire reference list. Do not add extra space between entries

9. Page numbers

- When citing a range of page numbers for an article, always give the complete numbers
- If an article continues on subsequent pages after being interrupted by other articles or advertisements, use a comma to separate the page numbers
- Use the abbreviation p. or pp. only with articles in newspapers, chapters in edited books, and articles from proceedings published as a book

10. Dates

- Follow the format year, month, day, with a comma after only the year: (2011, October 31).

Sample APA Reference List

	REFERENCES
Nonperiodical web document with no DOI	Centers for Disease Control and Prevention. (2010, June 1). <i>Teens behind the wheel: Graduated driver licensing</i> . Retrieved from http://www.cdc.gov/MotorVehicleSafety/Teen_Drivers/GDL/Teens_Behind_Wheel.html
Journal article, paginated by volume	Cumsille, P., Darling, N., & Martinez, M. L. (2010). Shading the truth: The pattern of adolescents' decisions to avoid issues, disclose, or lie to parents. <i>Journal of Adolescence</i> , 33, 285–296.
Online article with a DOI	Iemolo, F., Cavallaro, T., & Rizzuto, N. (2010). Atypical Alzheimer's disease: A case report. <i>Neurological Sciences</i> , 31, 643–646. doi:10.1007/s10072-010-0334-1
Chapter in an edited book	Jyonouchi, H. (2010). Possible impact of innate immunity in autism. In A. Chauhan, V. Chauhan, & W. T. Brown (Eds.), <i>Autism: Oxidative stress, inflammation, and immune abnormalities</i> (pp. 245–276). Boca Raton, FL: CRC Press.
Book in an edition other than the first	Quinn, G. R. (2010). <i>Behavioral science</i> (2nd ed.). New York, NY: McGraw-Hill Medical.
Online article, paginated by issue, with no DOI	Srivastava, R. K., & More, A. T. (2010). Some aesthetic considerations for over-the-counter (OTC) pharmaceutical products. <i>International Journal of Biotechnology</i> , 11(3–4), 267–283. Retrieved from http://www.inderscience.com

IEEE style

IEEE style consists of two elements: citations in the text and a reference list at the end of the document

IEEE Textual Citations

1. citations in the text are bracketed numbers, keyed to a numbered list of references that appears at the end of the document
 - Entries in the list are arranged in the order in which they are cited in the text and are numbered sequentially
 - Once a reference has been listed, the same number is used in all subsequent citations of that source
2. To cite references in the text, place the reference number or numbers immediately after the author's name, in square brackets, before any punctuation
3. Use et al. if there are three or more author names
4. You can also use the bracketed citation number or numbers as a noun.

The IEEE Reference List

1. Arranging entries

- Arrange the entries in the order in which they first are cited in the text, and then number them sequentially. Place the numbers in square brackets and set them flush left in a column of their own, separate from the body of the references. Place the entries in their own column, with no indents for turnovers

2. Authors

- List the author's first initial (or first and middle initials, separated by spaces), followed by the last name. In the case of multiple authors, use all names; use et al. after the first author's name only if the other names are not given. If an entry has an editor or translator in place of an author, add the abbreviation Ed. (or Eds. for editors) or Trans. following the name

3. Book titles

- Italicize titles of books. In English, capitalize the first word and all major words. In foreign languages, capitalize the first word of the title and subtitle, as well as any words that would be capitalized in that language

4. Publication information

- For books, give the city of publication, the country (if other than the United States), the publisher's name (abbreviated), and the year of publication. When two or more cities are given on a book's copyright page, include only the first. If the city is not well known, add the abbreviation for the name of the state or province (in Canada). If the publisher's name indicates the state, no state abbreviation is necessary

5. Periodical titles

- Italicize and abbreviate titles of periodicals. Capitalize all major words in the title

6. Article titles

- Place titles of print articles in quotation marks; do not use quotation marks for titles of articles found in electronic sources. Capitalize the first word of the title and subtitle. Do not capitalize the remaining words unless they are proper nouns

7. Electronic sources

- Follow the special style for electronic sources in which, most notably, the sequence of information is different from that for print material (the date follows the author, and the year comes before the month). Do not place article titles in quotation marks, and use periods rather than commas to separate sections. In addition to the basic information, give the medium and provide a way to locate the source by including, for example, a URL

8. Spacing

- Single-space the reference list, and do not add extra space between entries

9. Page numbers

- To give a page or a range of pages for a specific article in a book or periodical, use the abbreviation p. or pp. Write numbers in full

10. Dates

- For print sources, follow the format month (abbreviated), day, year (for example, Apr. 3, 2010 or Feb. 22–23, 2011). Do not abbreviate May, June, or July. For electronic sources, follow the format year, month (abbreviated), day (for example, 2011, Oct. 14)

Sample IEEE Reference List

Reference List	
Article in an online magazine	[1] S. Schmidt. (2008, June). Arthur C. Clarke, 1917–2008. <i>Analogsf.com</i> [Online]. Available: http://www.analogsf.com/0806/Obitclarke.shtml
Chapter in an edited book	[2] E. Castronova, "The changing meaning of play," in <i>Online Communication and Collaboration: A Reader</i> , H. M. Donelan, K. L. Kear, and M. Ramage, Eds. New York: Routledge, 2010, pp. 184–189.
Book in an edition other than the first	[3] L. Xinju, <i>Laser Technology</i> , 2nd ed. Boca Raton, FL: CRC Press, 2010, pp. 203–205.
Online journal article	[4] R. Marani and A. G. Perri. (2010). An electronic medical device for preventing and improving the assisted ventilation of intensive care unit patients. <i>Open Elect. Electron. Eng. J.</i> [Online]. 4, pp. 16–20. Available: http://www.benthamscience.com/open/toeej/openaccess2.htm
Standard	[5] <i>Testing and Evaluation Protocol for Spectroscopic Personal Radiation Detectors (SPRDs) for Homeland Security</i> , ANSI Standard T&E Protocol N42.48, 2010.
Journal article	[6] A. C. Mathieson, E. J. Hehre, C. J. Dawes, and C. D. Neefus, "An historical comparison of seaweed populations from Casco Bay, Maine," <i>Rhodora</i> , vol. 110, no. 941, pp. 1–10, 2008.

MLA Style (Modern Language Association)

MLA (Modern Language Association) style consists of two elements: citations in the text and a list of works cited at the end of the document

MLA Textual Citations

the textual citation typically includes the name of the source's author and the number of the page being referred to

The MLA List of Works Cited

1. Arranging entries

- Arrange the entries alphabetically by the author's last name. If two or more works are by the same author, arrange them alphabetically by title. Alphabetize works by an organization by the first significant word in the name of the organization

2. Book Titles

- Italicize titles of books and capitalize all major words. Note that in MLA style, prepositions are not capitalized.

3. Publication information

- Shorten the publisher's name where possible. For cities outside the United States, include the name of the province (in Canada) or country, abbreviated, unless the city is well known (such as Tokyo or London)

4. Periodical titles

- Italicize titles of periodicals and capitalize all major words. Omit any initial article

5. Article titles

- Place titles of articles and other short works in quotation marks and capitalize all major words

6. Electronic sources

- Include as much information as you can about electronic sources, such as author, date of publication, identifying numbers, and retrieval information.

7. Indenting

- Use a hanging indent, with the first line of each entry flush with the left margin and all subsequent lines indented one-half inch

8. Spacing

- Double-space the entire works-cited list. Do not add extra space between entries

9. Page numbers

- Do not use the abbreviation p. or pp. when giving page numbers. For a range of pages, give only the last two digits of the second number if the previous digits are identical (for example, 243–47, not 243–247 or 243–7). Use a plus sign (+) to indicate that an article continues on subsequent pages, interrupted by other articles or advertisements

10. Dates

- Follow the format day month year, with no commas (for example, 20 Feb. 2009). Spell out May, June, and July; abbreviate all other months (except Sept.) using the first three letters followed by a period

The MLA List of Works Cited

1. Medium

- With a few exceptions explained below, list the medium of publication, followed by a period, as the last part of any entry. Examples include Print, Web, Radio, Television, CD, CD-ROM, Audiocassette, Film, Videocassette, DVD, Performance, Address, MS (for manuscript), TS (for typescript), E-mail, PDF file, Microsoft Word file, JPEG file, MP3 file

Sample MLA List of Works Cited

Works Cited	
Geary, Patrick, ed. <i>Readings in Medieval History</i> . 4th ed. North York, ON: U of Toronto P, 2010. Print.	Book in an edition other than the first
Kunnan, Anthony J. "Publishing in the Era of Online Technologies." <i>Modern Language Journal</i> 94.4 (2010): 643-45. <i>Academic OneFile</i> . Web. 12 Feb. 2011.	Article from a database
Marx, Karl. "Proletarians and Communists." <i>Marx Today: Selected Works and Recent Debates</i> . Ed. John F. Sitton. New York: Palgrave-Macmillan, 2010. 51-56. Print.	Chapter in an edited book
Melia, Mike. "Atlantic Garbage Patch: Pacific Gyre Is Not Alone." <i>Huffington Post</i> . HuffingtonPost.com, 15 Apr. 2010. Web. 13 Feb. 2011.	Article in an online newspaper
Mooney, William. "Sex, Booze, and the Code: Four Versions of the <i>Maltese Falcon</i> ." <i>Literature-Film Quarterly</i> 39.1 (2011): 54-72. Print.	Journal article

Comparison

Aspect	APA Style	IEEE Style	MLA Style
Basic Citation Type	Author-Date system (e.g., <i>Smith, 2010</i>)	Numbered system (e.g., <i>[1]</i>)	Author-Page system (e.g., <i>(Smith 25)</i>)
In-text Citation Format	(Author, Year) — e.g., (American Nurses Association, 2010)	[Number] — e.g., [5], [8]–[10]	(Author Page) — e.g., (Hawking 45)
Organization as Author	Use organization name — e.g., (American Nurses Association, 2010)	Cited by reference number — e.g., [3]	Use organization name — e.g., (World Health Organization 12)

Aspect	APA Style	IEEE Style	MLA Style
Abbreviated Organization Name	Use abbreviation after first full citation — e.g., (International Business Machines [IBM], 2011); later (IBM, 2011)	Not applicable (number system)	Use abbreviation after first full citation, if needed
Unknown Author	Use shortened title — e.g., (“World Scientists,” 2009)	Not applicable (numbered citation refers to reference list)	Use title of work in quotation marks — e.g., (“World Scientists” 2009)
Anonymous Author	Treat “Anonymous” as real name — e.g., (Anonymous, 2011)	Not applicable	Rarely used; generally use title instead
Multiple Authors (Same Last Name)	Include initials — e.g., B. Porter (2012), A. L. Porter (2012)	Not applicable	Include first names or initials to differentiate
Multiple Sources in One Citation	Alphabetical order; separate by semicolons — e.g., (Houlding, 2011; Jessen, 2010)	Sequential numbers — e.g., [5]; [8]–[10]	Alphabetical order — e.g., (Houlding 45; Jessen 60)
Personal Communication	Include phrase “personal communication” and date — e.g., D. E. Walls (personal communication, April 3, 2011)	Rarely cited; if used, just [number] linked to reference list	Mention in text; not usually cited in works cited
Electronic Documents	Cite like print; if no author, use title; if no date, use <i>n.d.</i> (e.g., Hoffman, n.d.)	Include author, date (Year, Mon. Day), medium, and URL — e.g., [Online]. Available: URL	Include author, title, publisher, date, medium (Web), retrieval date, and URL (if necessary)
Page / Paragraph Numbers	(Tong, 2010, para. 4) or (The Reality section, para. 3)	Use <i>p.</i> or <i>pp.</i> for pages — e.g., pp. 152–159	No <i>p.</i> or <i>pp.</i> ; use 243–47, 243+ for continued pages
Arrangement of References	Alphabetical by author’s last name; same author works	Numerical order (by first citation in text)	Alphabetical by author’s last name

Aspect	APA Style	IEEE Style	MLA Style
	by date (earliest to latest)		
Book Titles	<i>Italicized</i> ; capitalize first word, subtitle's first word, and proper nouns	<i>Italicized</i> ; capitalize first and all major words	<i>Italicized</i> ; capitalize all major words (except prepositions)
Periodical Titles	<i>Italicized</i> ; capitalize all major words	<i>Italicized</i> and abbreviated; capitalize all major words	<i>Italicized</i> ; capitalize all major words; omit initial article
Article Titles	Not italicized or quoted; capitalize only first word and proper nouns	"Quoted" for print; not quoted for online; capitalize first word	"Quoted"; capitalize all major words
Publication Information (Books)	Include publisher (abbreviated), city, and state/country	Include city, publisher (abbreviated), year	Include city and publisher (shortened form)
Electronic Sources (Details)	Include DOI if available; retrieval date if content may change	Include medium, date, and URL; format: Year, Mon. Day	Include sponsor/publisher, date, medium, retrieval date, and URL (if needed)
Indenting	Hanging indent: 1st line flush left, others indented 0.5 inch	No indent; numbers flush left, text in column	Hanging indent: 1st line flush left, others indented 0.5 inch
Spacing	Double-spaced; no extra space between entries	Single-spaced; no extra space between entries	Double-spaced; no extra space between entries
Page Number Format	Always give full range (121–124)	Use <i>p.</i> or <i>pp.</i> ; full numbers (152–159)	Do not use <i>p.</i> or <i>pp.</i> ; shorten second number (243–47)
Date Format	Year, Month Day — e.g., (2011, October 31)	Month (abbr.) Day, Year — e.g., Apr. 3, 2010; or Year, Mon. (abbr.) Day for online	Day Month Year — e.g., 20 Feb. 2009

Aspect	APA Style	IEEE Style	MLA Style
Medium Indication	Not required	Indicate medium for electronic sources (e.g., [Online])	Required — e.g., Print, Web, DVD, E-mail, PDF file
Example Citation (Book)	Sokolova, G. N. (2010). <i>Economic stratification in Belarus and Russia: An experiment in comparative analysis. Sociological Research</i> , 49(3), 25–26.	[1] G. N. Sokolova, <i>Economic Stratification in Belarus and Russia: An Experiment in Comparative Analysis</i> , Sociol. Res., vol. 49, no. 3, pp. 25–26, 2010.	Sokolova, Galina N. “Economic Stratification in Belarus and Russia: An Experiment in Comparative Analysis.” <i>Sociological Research</i> , vol. 49, no. 3, 2010, pp. 25–26. Print.

Ethical Considerations in Technical Writing

The Core of Ethical Technical Writing

Ethical technical writing is about doing the right thing. It's about being honest, transparent, and responsible in all your communications

Key Principles

1. Honesty - Don't exaggerate claims or misrepresent data
2. Accuracy - Ensure all facts, figures, and technical details are correct
3. Clarity - Write in a way that avoids misinterpretation, especially in safety instructions
4. Completeness - Provide all necessary information, not just what makes your product look good

Ethical Considerations in Technical Writing

Intellectual Property (IP) - refers to creations of the mind

- Copyright is one of the main forms of IP

Other Types of IP :

1. Patents - Protect inventions
2. Trademarks - Protect brand names, logos, and slogans
3. Trade Secrets - Protect confidential business information\

Copyright

What's protected by copyright?

1. Original Expression
2. Code
3. Illustrations and Diagrams
4. Reports and Documents

What's NOT protected?

1. Ideas and Concepts
2. Facts and Data

Copyright Violations and Plagiarism

How to avoid plagiarism:

1. Proper Citation - Always credit your sources
2. Paraphrasing - Rewrite the information in your own words and style
3. Quoting - Use quotation marks for direct quotes and cite the source immediately
4. Permission - For copyrighted material you may need to get explicit permission from the owner

Plagiarism and Academic Integrity

Academic integrity is a commitment to five fundamental values: honesty, trust, fairness, respect, and responsibility

It means:

1. Doing your own work
2. Accurately reporting data, even if it's not the result you hoped for
3. Giving credit where credit is due

Plagiarism can take many forms:

1. Direct - Copying text word-for-word without quotation marks and a citation
2. Paraphrasing - Rewriting someone's ideas in your own words without citing the original source
3. Self - Reusing your own previously submitted work without permission from your instructor
4. Mosaic - Patching together phrases and sentences from different sources without proper citation

Why Does This Matter for Engineers?

1. Safety Risks
2. Loss of Trust
3. Legal Action

Ethical Representation of Data and Results

- Your writing must be an honest and accurate reflection of your work

The Impact of Unethical Data Reporting

1. Project Failure
2. Loss of Credibility
3. Legal Consequences
4. Safety Risks

Upholding Confidentiality

1. Think "Need-to-Know" - Share confidential information only with colleagues who absolutely need it to do their job.
2. Secure Your Work - Lock your computer when you walk away.
3. Professional Discretion - Be mindful of what you post online.
4. A simple rule - If you're unsure whether something is confidential, treat it as if it is. When in doubt, ask your supervisor

Version Control

- Version control is a system that records changes to a file or set of files over time so you can recall specific versions later.

For engineers, it's crucial for:

1. Collaboration
2. Tracking
3. Backup

Feature	Centralized Version Control Systems (CVCS)	Distributed Version Control Systems (DVCS)
How it works	A single, central server stores all versions. Developers "check out" files, make changes, and then "check in" the new version.	Every developer has a full copy of the entire project history on their local machine. Changes are shared between these local repositories.
Pros 👍	Simpler to set up.	No single point of failure; you can work offline.
Cons 👎	Single point of failure. If the central server is down, no one can save their changes.	Can have a steeper learning curve initially.
Examples	CVS, Subversion (SVN).	Git (the most popular), Mercurial.

Key Version Control Concepts

1. Repository (Repo): A database that contains all the files and the complete history of a project.
2. Commit: A snapshot of your repository at a specific point in time.
3. Branching: A parallel version of the repository.
4. Merging: The process of combining changes from one branch into another
5. Conflict: This happens when two different branches change the same line of code or part of a file, and the version control system can't automatically decide which change to keep.

Introduction to Git

Git is a powerful and widely-used Distributed Version Control System. It was created by Linus Torvalds, the creator of Linux.

Why Git?

1. Speed
2. Flexibility
3. Community
4. Industry Standards

Best Practices for Engineers

1. Commit frequently and with purpose
2. Write clear commit messages
3. Use branches for everything
4. Resolve conflicts promptly
5. Leverage remote repositories

Types of Collaborative Tools

1. Document-Centric Tools - Google Docs, Microsoft Word, Confluence
2. Communication Tools - Slack/Microsoft Teams, Zoom/Google Meet, email
3. Project Management Tools - Trello, Jira, Asana
4. Version Control Systems - Git, SVN